

Robert Leggieri

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Objective

Highly motivated IT professional with 20 years of professional experience eager to transition into a data-centric role. Seeking to apply a robust technical foundation and newly acquired academic knowledge in data science to effectively analyze complex datasets, develop predictive models, and provide actionable insights that drive organizational success.

Skills

AI/ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, CNNs, Hugging Face Transformers

Programming & Scripting: Python (NumPy, Pandas, Matplotlib, Seaborn, OpenCV, NLTK), Excel VBA, PowerShell

Data Processing: XSLT, JScript, data curation, workflow automation, ETL pipelines

Analytical, detail oriented, independent, efficient, problem-solver, confident in communication, enthusiastic to learn

Projects

MSAI NLP Final Project: Debiasing LLMs Fine-Tuned for NLI Task

August 2024 – December 2024

- Implemented a novel loss function informed by dataset cartography to mitigate dataset artifacts in the MNLI dataset.
- Achieved improved model generalization on adversarial and out-of-domain examples, demonstrating a potential path toward bias mitigation in large language models.

Data Science Capstone: Deep Learning Models for Lung Cancer Prediction

December 2023 – March 2024

- Developed CNN and Transfer Learning models using TensorFlow to classify lung cancer pathologies in CT scans.
- Leveraged interpretability techniques (Guided Backpropagation, Grad-CAM) to enhance transparency in model decision-making, addressing ethical concerns in medical AI.
- Demonstrated expertise in data exploration and statistical analysis of large, complex image datasets.

Custom Reports for LandXML Data Processing

Ongoing

- Automated data curation and report generation via XSLT and JScript for Autodesk Civil 3D LandXML exports.
- Reduced manual processing time by 80% through data transformation and workflow optimization.
- Showcased proficiency in data management best practices, reproducibility, and version control.

Data Conversion: Payroll & Accounting Migration to BQE Core

February 2022 – April 2022

- Oversaw the migration of 30+ years of legacy payroll and accounting data using Python-based ETL pipelines.
- Ensured data integrity through comprehensive mapping, cleansing, and validation processes.
- Demonstrated competency in large-scale data handling and cross-platform integration.

Experience

Network Manager, Gutschick, Little & Weber, P.A. – Burtonsville, MD

July 2004 – Present

- Provide strategic oversight of network infrastructure while emphasizing automation, security, and operational efficiency for over 40 users.
- Utilize PowerShell and Python scripting to streamline workflows and integrative solutions, ensuring scalability.
- Collaborate with diverse teams to optimize resource allocation, reducing operating costs through data-driven insights.
- Maintain awareness of evolving data management solutions, ensuring alignment with organizational objectives.

Education

University of Texas at Austin – M.S., Artificial Intelligence, GPA: 4.0

Expected Graduation May 2026

Coursework in Deep Learning, Natural Language Processing, Ethics, Online Learning and Optimization, AI in Healthcare

University of Maryland Global Campus – B.S., Data Science, *Summa Cum Laude*

May 2024

Coursework in Machine Learning, Artificial Intelligence, Data Ethics, Python, Statistics